

MinVol three-dimensional contact bound

Repository source: README.md

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Public-package construction checkpoint - not yet released

This fresh-history repository is being prepared for a computer-assisted preprint and exact reproducibility package for the proposed universal bound

$$\text{Vol}(K) \geq \frac{26220940089713\pi}{200000000000000} d^3 = 0.41187756378030227424817892598 \dots d^3$$

for every convex body K in $\mathbb{R}^3$ of constant width d .

The current checkpoint contains a standalone exact certificate, independent Python and base-R audits, adversarial rejection tests, and a lean Paper v1 manuscript reconciled automatically against the machine-readable certificate. It is still an unreleased research draft. Final authorship, affiliation, citation, license, repository, tag, and release metadata remain open.

Paper

- [Reference PDF](#)
- [LaTeX source](#)
- [Paper build and reconciliation guide](#)
- [Rendered paper-guide PDF](#)

The reference manuscript states the proof architecture and claim limitations in nine pages. Its certificate-critical constants and lower-branch partition are generated from the exact JSON rather than copied manually.

Reviewer path

1. Read the [paper PDF](#).
2. Read the [certificate status](#) and [certificate guide](#).
3. Verify the checksum manifests.
4. Run the exact and independent Python paths with `./run_python_checks.sh`.
5. With base R installed, run the complete replay with `./run_all.sh`.
6. Build the paper and Markdown documentation in a clean environment.

Replay

Install the pinned Python dependency:

```
python3 -m pip install -r requirements.txt
```

Run the exact verifier, binary64 audit, ten reconstruction tests, seven adversarial mutation tests, exact-constants and release-document checks, and paper/certificate reconciliation:

```
./run_python_checks.sh
```

Expected final markers include:

```
MINVOL UNIVERSAL CONTACT-BOUND CERTIFICATE: EXACT
MINVOL UNIVERSAL CONTACT-BOUND BINARY64 AUDIT: PASS
MINVOL ADVERSARIAL MUTATION SUITE: PASS
MINVOL EXACT-CONSTANTS APPENDIX: CURRENT
MINVOL RELEASE-DOCUMENT CHECK: PASS
MINVOL PAPER-CERTIFICATE RECONCILIATION: PASS
```

With base R installed, run all language paths:

```
./run_all.sh
```

To refresh the archived base-R transcript deliberately:

```
./run_r_checks.sh 2>&1 | tee certificate/transcripts/base_r_audit.txt
```

After changing a protected source, transcript, or reference artifact, refresh and verify the checksum manifests:

```
python3 refresh_checksums.py
```

```
python3 refresh_checksums.py --check
```

Build documentation

Build the paper without overwriting its committed reference PDF:

```
./build_paper.sh
```

Build local PDFs for every Markdown source under `build/markdown/`:

```
./build_markdown_pdfs.sh
```

The committed Markdown/PDF pairs are under `rendered/markdown/`. Their pairing and PDF preflight checks are run by continuous integration and can also be run locally.

Proof architecture

The proof splits the normalized circumradius interval at

$$R_* = \frac{305695601}{500000000} = 0.611391202.$$

Below the split, a gap-free exact radial-propagation certificate has unique bottleneck 03B02. Above the split, four balanced circumsphere contacts force a 98,306-ray polyhedral core. An exact determinant transfer, a dominant-edge coordinate-width estimate, and three optimized antipodal cap pairs produce a near-Jung lower bound strictly above the bottleneck. Exact integer and rational arithmetic supplies the finite proof authority; binary64 Python and base R are independent audits.

Documentation

The editable sources are Markdown. Checksummed reference PDFs mirror the same repository-relative tree under `rendered/markdown/`.

Status, release, and provenance

- Verification status: [Markdown](#) | [PDF](#)
- Reproducibility guide: [Markdown](#) | [PDF](#)
- Clean-source record: [Markdown](#) | [PDF](#)
- AI assistance and provenance: [Markdown](#) | [PDF](#)
- Draft release notes: [Markdown](#) | [PDF](#)
- Release checklist: [Markdown](#) | [PDF](#)

Certificate documentation

- Certificate specification: [Markdown](#) | [PDF](#)
- Trust boundary: [Markdown](#) | [PDF](#)
- Claims and limitations: [Markdown](#) | [PDF](#)
- Independent-audit record: [Markdown](#) | [PDF](#)
- Certificate guide: [Markdown](#) | [PDF](#)
- Certificate status: [Markdown](#) | [PDF](#)

Mathematical proof documentation

- Mathematical overview: [Markdown](#) | [PDF](#)
- Claim dependencies: [Markdown](#) | [PDF](#)
- Proof ledger: [Markdown](#) | [PDF](#)
- Internal proof audit: [Markdown](#) | [PDF](#)
- Source ledger: [Markdown](#) | [PDF](#)
- Appendix A, exact constants: [Markdown](#) | [PDF](#)
- Appendix B, certificate architecture: [Markdown](#) | [PDF](#)
- Appendix C, replay transcripts: [Markdown](#) | [PDF](#)
- Paper guide: [Markdown](#) | [PDF](#)
- This README: [Markdown](#) | [PDF](#)

Scope

The package proposes a universal lower bound. It does not identify a minimizing body, prove sharpness or equality, or prove the Meissner conjecture. The Meissner value is an explicit-body comparison value, not a universal minimum proved here. The result has not yet been externally reproduced, peer reviewed, or publicly released.

The earlier public MinVol spectral announcement remains a separate repository: [swihart/minvol-degree3-spectral-bound](#). Further universal refinements are under investigation but are outside this v1 package.